Advantages and Shortcomings of Each of the Systems

While the Viscosity-Graded Specification represented an improvement over the Penetration-Graded Specification, it also had some of the same limitations. Viscosity measurements at 275°F (135°C) allowed the user to determine the stiffness of the asphalt binder at typical mixing temperatures in an HMA mixing facility. Viscosity measurements at 140°F (60°C) allowed the user to determine the stiffness of the asphalt binder at typical high pavement temperatures. Despite the addition of intermediate temperature measurements (such as penetration) to some specifications, the stiffness of the asphalt binder at cold temperatures could only be inferred from the intermediate temperature data. As in Figure 2.14, Figure 2.15 illustrates that the actual performance could be significantly different for asphalt binders with the same grade.

As with the Penetration-Graded Specification, short-term aging effects were captured through the use of the TFOT or RTFOT, but there was no provision to conduct long-term aging of asphalt binders to simulate what happens to the asphalt stiffness after being in a pavement for a number of years.

Finally, although viscosity is a fundamental property, the viscosity test is only useful in measuring the viscous component of an asphalt binder. This is not an issue for conventional, refined asphalt binders but becomes a greater issue when using asphalt binders produced using polymer modification. For these asphalt binders, the elastic component of the asphalt binder really is not properly considered when using the viscosity test. As such, the viscosity of a modified asphalt binder may not provide sufficient information about the asphalt binder's physical properties to properly judge its performance potential.

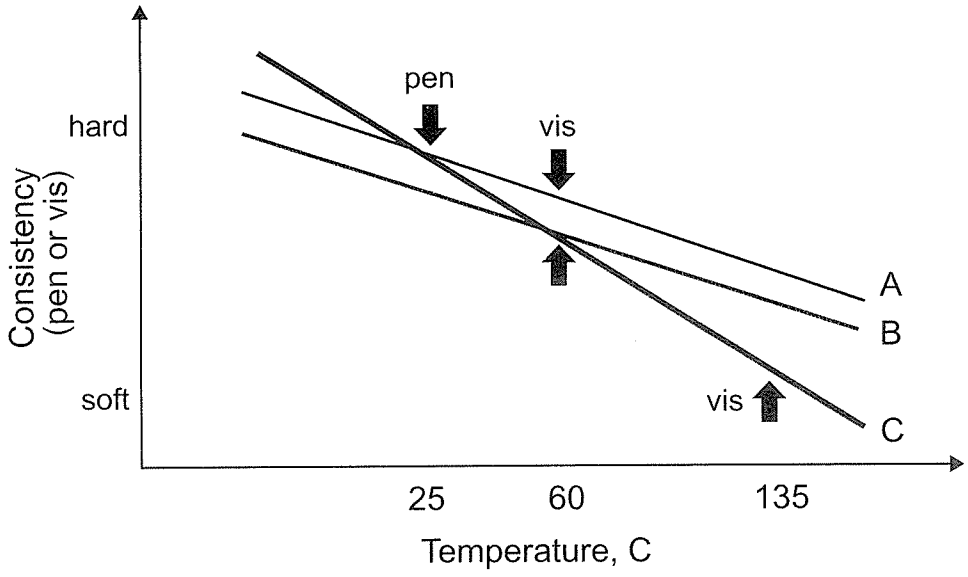


Figure 2.15 Asphalt Binder Properties—Viscosity Grading

In addition to the principal tests (penetration or viscosity) for defining grade, several other tests are required to determine specific properties for these products. For general reference, the tests are listed in Table 2.2, along with the appropriate AASHTO or ASTM test method and indications of which specification requires the test. Always refer to the current AASHTO or ASTM specification.

As a final note, since viscosity is a fundamental material property (unlike penetration), there is no universal relationship between the two properties. The relationship between penetration and viscosity varies for different asphalts produced from different crude sources.

Table 2.2
Asphalt Cement Specification Tests

	Test	Test Method		Specification[3]	
		AASHTO	ASTM	Viscosity	Penetration
Unaged Asphalt	Viscosity @ 140°F (60°C)	T 202	D 2171	x	
	Viscosity @ 275°F (135°C)	T 201	D 2170	x	
	Penetration @ 77°F (25°C)	T 49	D 5	x	x
	Flash Point	T 48	D 92	x	x
	Solubility in TCE	T 44	D 2042	x	x
	Ductility @ 77°F (25°C)	T 5	D 113		x
	Spot Test[1]	T 102	n/a	x	x
Aged Residue	Mass Loss[2]	T 179 T 240	D 1754 D 2872	x x	x
	Viscosity @ 140°F (60°C)	T 202	D 2171	x	
	Viscosity @ 275°F (135°C)	T 201	D 2170	x	
	Penetration @ 77°F (25°C)	T 49	D 5	x	x
	Ductility @ 77°F (25°C)	T 5	D 113	x	x

[1] Optional test in AASHTO specification
[2] Mass Loss is conducted by either TFO or RTFO (TFO listed first). Residue is obtained after aging.
[3] Viscosity specifications are AASHTO M 226 and ASTM D 3381. Penetration specifications are AASHTO M 20 and ASTM D 946. Note that the viscosity specification includes three grading tables for which test requirements may differ from what is shown in this table.

2.3 Dominant Practice: Performance Graded (PG) Asphalt Binder

2.3.1 Introduction

Recognizing the deficiencies in the viscosity and penetration grading systems, state highway agencies decided to fund a research program to develop a new system for specifying paving asphalt. In 1987, the Strategic Highway Research Program (SHRP) began developing new, performance-related tests for measuring the physical properties of asphalt binders. A major result of this $50 million research effort was the Performance Graded Asphalt Binder specification. Because the specification was to be used equally with modified as well as unmodified asphalts, the term *asphalt binder* was substituted for *asphalt cement.*

2.3.2 Asphalt Behavior

Because of its viscoelastic nature, asphalt binder behavior depends on both temperature and rate of loading. As shown in Figure 2.16, the amount of flow for an asphalt binder could be the same for one hour at 60°C or 10 hours at 25°C. In other words, the effects of time and temperature are related in the measurement of flow properties of an asphalt binder. Thus, the flow of an asphalt binder at high temperature loaded in a short period may be equivalent to the flow occurring at a lower temperature for a longer loading duration. This useful concept is known as time-temperature superposition and it applies to linear viscoelastic materials.

High Temperature

In hot conditions, such as summer temperatures, or when subjected to sustained loads, such as slow-moving trucks, an asphalt binder acts like a viscous liquid. Viscosity is the physical material characteristic used to describe the resistance of liquids to flow.

Asphalt binders typically display either Newtonian or non-Newtonian behavior as shear rate is changed. Newtonian fluids have a linear relationship between resisting force (shear stress) and relative velocity (rate of shear strain). In other words, if you apply twice the force to a Newtonian fluid, it will move twice as fast. Air, water, and hot asphalt (at temperatures greater than 60°C) are common Newtonian fluids. For these materials, viscosity (μ) is constant regardless of shear rate as shown in Figure 2.17.

Asphalt binders, particularly modified asphalts, may exhibit non-Newtonian behavior. Some asphalt materials may exhibit pseudoplastic or "shear thinning" behavior. This type of behavior, demonstrated in paints, is characterized

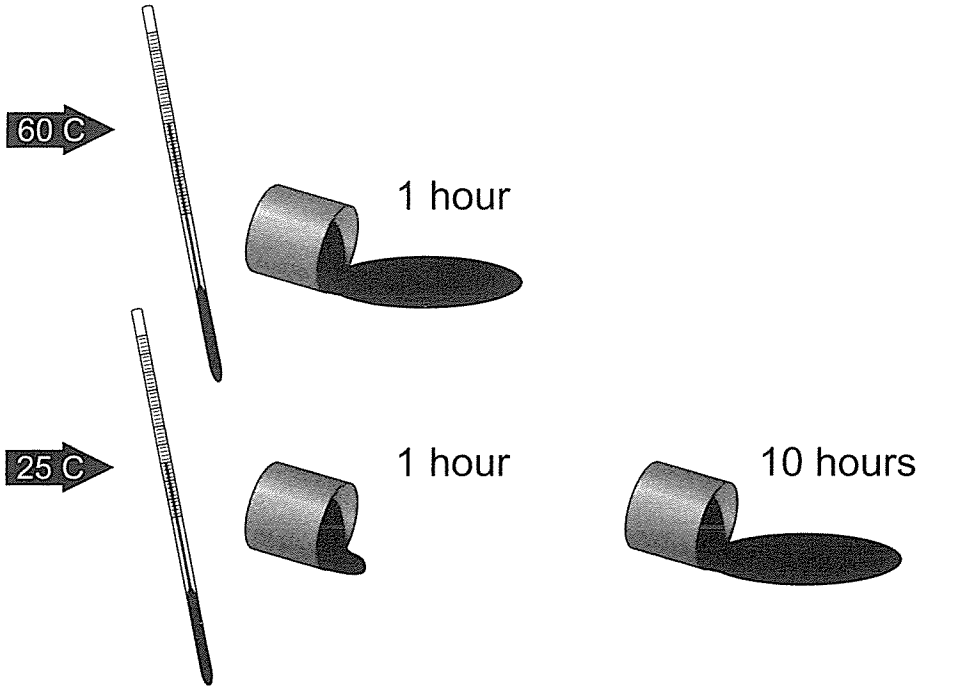


Figure 2.16 Time-Temperature Superposition

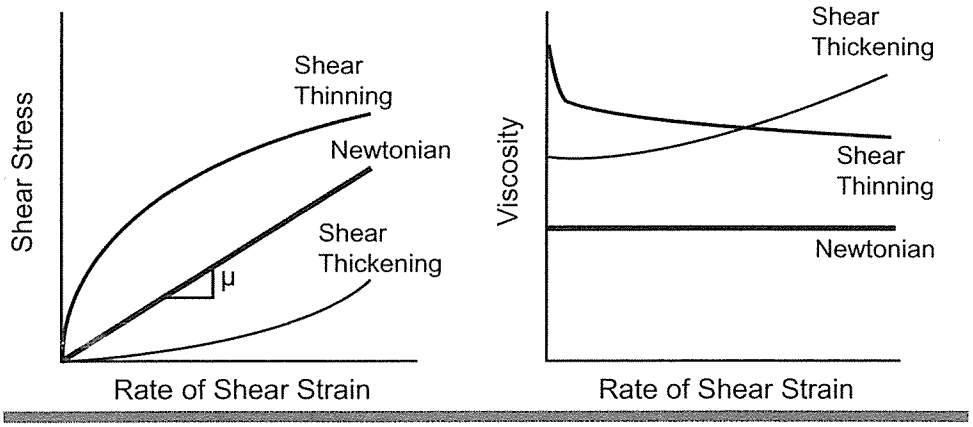


Figure 2.17 Behavior of Asphalt Binders as Shear Rate Changes

by a decrease in viscosity as shear rate is increased (see Figure 2.17). In other words, the faster one stirs the material, the thinner (more fluid) it gets. At moderate temperatures, asphalt exhibits "shear thinning" behaviors since its viscosity decreases at increasing shear rates.

Another type of non-Newtonian behavior, although less common, is dilatant or "shear thickening" behavior. This type of behavior, demonstrated in clay slurries, is characterized by an increase in viscosity as shear rate is increased (see Figure 2.17). In other words, the faster one stirs the material, the thicker (less fluid) it gets.

Low Temperature

In cold conditions such as winter temperatures, or when subjected to rapidly applied loads such as fast-moving trucks, an asphalt binder behaves like an elastic solid. Elastic solids are like rubber bands; when loaded they deform, and when unloaded, they return to their original shape.

If stressed beyond the material capacity or strength, elastic solids may break. Even though asphalt binders are elastic solids at low temperatures, they may become too brittle and crack when loaded. For this reason, low-temperature cracking sometimes occurs in asphalt pavements during cold weather. In these cases, the load is applied not by vehicles but by internal stresses that accumulate in the asphalt pavement as a result of low temperatures. As the temperature decreases, the asphalt binder cementing the aggregates shrinks, leading to a buildup of stresses. If the internal stresses in the asphalt binder exceed the binder's strength, then the interface between the aggregates will break, resulting in the formation of a crack.

Intermediate Temperature

At intermediate temperatures, asphalt binders exhibit the characteristics of both viscous liquids and elastic solids. When heated, asphalt acts as a lubricant, allowing the aggregate to be mixed, coated, and tightly compacted to form a smooth, dense surface. After cooling, the asphalt acts as the glue to hold the aggregate together in a solid matrix. In this finished state, the behavior of the asphalt is *viscoelastic,* meaning it has both elastic and viscous characteristics, depending on the temperature and rate of loading.

Aging

Because asphalt binders are composed of organic molecules, they react with oxygen from the environment. This reaction is called oxidation and it changes the structure and composition of asphalt molecules. Oxidation causes the asphalt binder to become more brittle, leading to an effect called oxidative hardening. Oxidative hardening happens at a relatively slow rate in a pavement, although it occurs faster in warmer climates and during warmer seasons. Because of this hardening, old asphalt pavements may be more susceptible to cracking than newer pavements. However, it should be noted that improperly compacted asphalt pavements might exhibit premature oxidative hardening. In this case, inadequate compaction leaves a higher percentage of interconnected air voids, which allows more air to penetrate the asphalt mixture, leading to more oxidative hardening early in the life of the pavement.

In practice, a considerable amount of oxidative hardening occurs before an asphalt mixture is placed. At the hot mixing facility, asphalt binder is added to the hot aggregate and the mixture is maintained at elevated temperatures for a time. Because the asphalt cement exists in thin films covering the aggregate, the oxidation reaction may occur at a relatively fast rate.

Other forms of hardening include volatilization and physical hardening. Volatilization occurs during hot mixing and construction, when volatile

components tend to evaporate from the asphalt binder. Physical hardening occurs when asphalt binders have been exposed to low temperatures for long periods. When the temperature stabilizes at a constant low value, the asphalt cement continues to shrink and harden. Physical hardening is more pronounced at temperatures less than 32°F (0°C) and must be considered when testing asphalt binders at very low temperatures.

2.3.3 PG Asphalt Binder Tests

Rotational Viscometer

A rotational viscosity test is used to determine the flow characteristics of an asphalt binder at a high temperature to provide some assurance that it can be pumped and handled at the hot-mixing facility. A rotational coaxial cylinder viscometer as shown in Figure 2.18 and described in AASHTO T 316 is necessary to evaluate the various types of asphalt binders. Unlike capillary tube viscometers, rotational viscometers have larger clearances between the components, allowing for better applicability to both modified and unmodified asphalts.

When a rotational viscometer is used, viscosity is determined by measuring the torque required to maintain a constant rotational speed of a cylindrical spindle that is submerged in an asphalt binder sample at

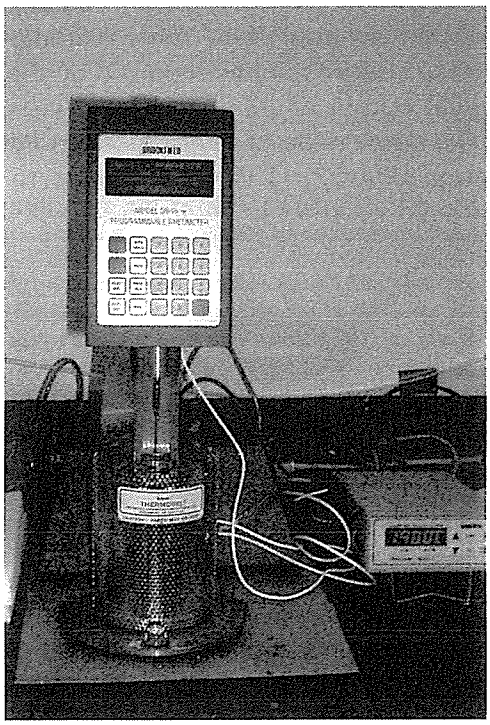


Figure 2.18 Rotational Viscosity Equipment

a constant temperature (see Figure 2.19). This torque is directly related to the asphalt binder viscosity, which is calculated automatically by the viscometer. Because this binder viscosity is used to ensure that the asphalt is fluid enough at normal operating temperatures to pump and mix with aggregate, it is measured on original or "tank" asphalt. The viscometer can also be used to develop temperature-viscosity charts for estimating mixing and compaction temperatures for use in mixture design.

In the rotational viscosity test, a sample of asphalt binder is weighed into a preheated sample chamber. The amount of asphalt used is typically 8 to 11 grams and varies with the size of spindle. The sample chamber containing the binder sample is placed in the preheated thermo-container operating at the proper test temperature (usually 275°F/135°C), the preheated spindle is lowered into the sample, and the binder is ready to test when the temperature stabilizes, usually within 30 minutes.

At the end of this time, the rotational viscosity test is started by activating the motor so that the spindle turns at a prescribed rate—20 rpm for specification testing. As the spindle rotates, the torque in the torsional spring is measured and converted to viscosity. A waiting period of 15 minutes may be needed to reach a uniform sample temperature. During this period with the spindle rotating, the viscosity reading and the percentage of torque can be observed on the digital display. If the percentage of torque is less than 2 percent or greater than 98 percent, then a different-size spindle will be required. As the temperature equalizes, the viscosity reading will stabilize and test results are recorded. Three viscosity readings are recorded at 1-minute intervals and the average of the three readings is reported.

Users are cautioned to check the units reported by the rotational viscometer because the digital output of some rotational viscometers is in units of centipoise (cP). The conversion of centipoises to Pascal-seconds, Pa•s is given by the equation 1000 cP = 1 Pa•s. Therefore, to obtain viscosity in Pascal-seconds, rotational viscosity in centipoises is divided by 1000.

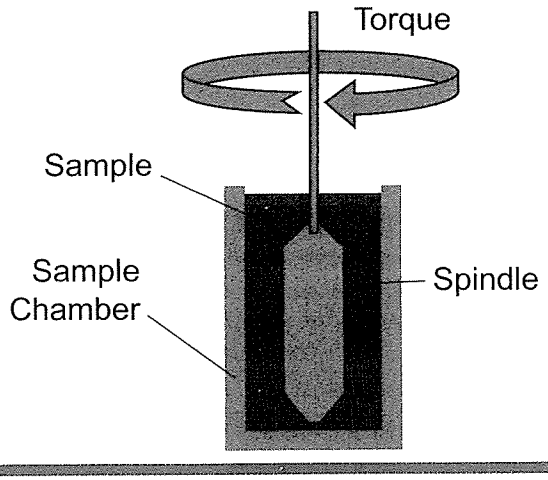


Figure 2.19 Rotational Viscosity Operation

Finally, in addition to testing at 135°C for specification purposes, it may be desirable to determine the asphalt binder viscosity at additional temperatures. For instance, many asphalt technologists use equiviscous temperatures for mixing and compaction during mix design. To determine these temperatures, a viscosity-temperature relationship for the asphalt binder is needed, meaning that the viscosity at an additional—usually higher—temperature, such as 163°C, is needed. The rotational viscosity test is conducted as before but at the new test temperature.

Dynamic Shear Rheometer (DSR)

The dynamic shear rheometer is used to test asphalt binders and measure their rheological properties, including complex shear modulus (G^*, pronounced "G-star") and phase angle (δ, the Greek letter "delta"), at intermediate to high test temperatures. These parameters can be used to characterize both viscous and elastic behavior of asphalt binders. The complex shear modulus, G^*, is a measure of the total resistance of a material to deformation when exposed to a sinusoidal shear stress load. G^* consists of both elastic (recoverable) and viscous (non-recoverable) components. The phase angle, δ, is an indicator of the relative amounts of viscous and elastic components. The values of G^* and δ for asphalt binders are highly dependent on the test temperature and frequency of loading. At high temperatures, asphalt binders behave like viscous fluids with little capacity for recovering or rebounding. In this case, the asphalt binder could be represented by the vertical axis (viscous component only) in Figure 2.20; there would be no elastic component of G^*, since $\delta = 90°$. At very low temperatures, asphalts behave like elastic solids, which rebound from deformation completely. This condition is represented by the horizontal axis (elastic component only) in Figure 2.20. In this case, there is no viscous component of G^*, since $\delta = 0°$.

Under normal pavement temperatures and traffic loadings, asphalt binders act with the characteristics of both viscous liquids and elastic solids and are said to be "viscoelastic" materials. By measuring G^* and δ, the DSR provides a complete picture of the behavior of asphalt at a range of pavement service temperatures. The vector arrows in Figure 2.20, G^*_1 and G^*_2, represent the complex modulus values of Binders 1 and 2. When these asphalt binders are loaded, part of their deformation is elastic *(E)* and part is viscous *(V)*. Even though both asphalt binders in Figure 2.20 are viscoelastic and have the same G^*, Binder 2 is more elastic than Binder 1, because of its smaller δ. Because Binder 2 has a larger elastic component, it will recover much more deformation from an applied load than Binder 1. This example clearly shows that G^* alone cannot describe asphalt behavior; the value of δ is also needed.

The operation of a DSR is fairly basic. A sample of asphalt binder is "sandwiched" between two parallel plates, one that is fixed and one that oscillates (see Figure 2.21). As the plate oscillates, the centerline of the plate at point A (indicated by the dark vertical line) moves to point B. From point B, the plate centerline moves back and passes point A to

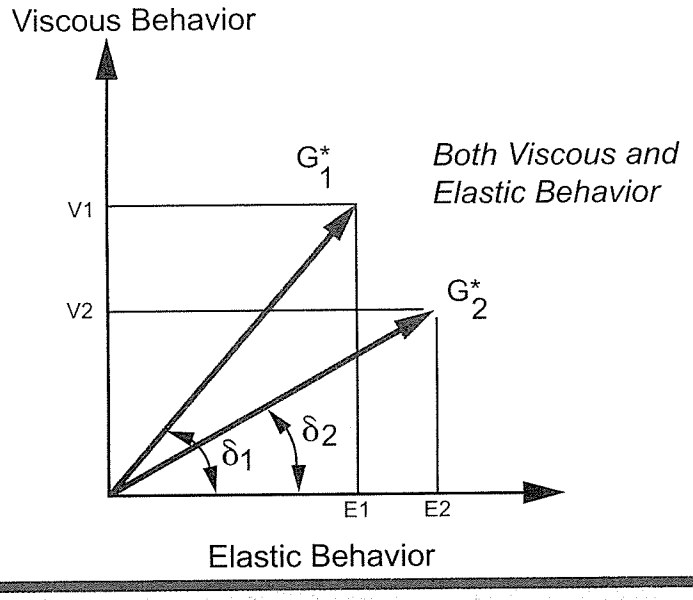


Figure 2.20 Graphical Representation of Shear Modulus (G*) and Phase Angle (δ)

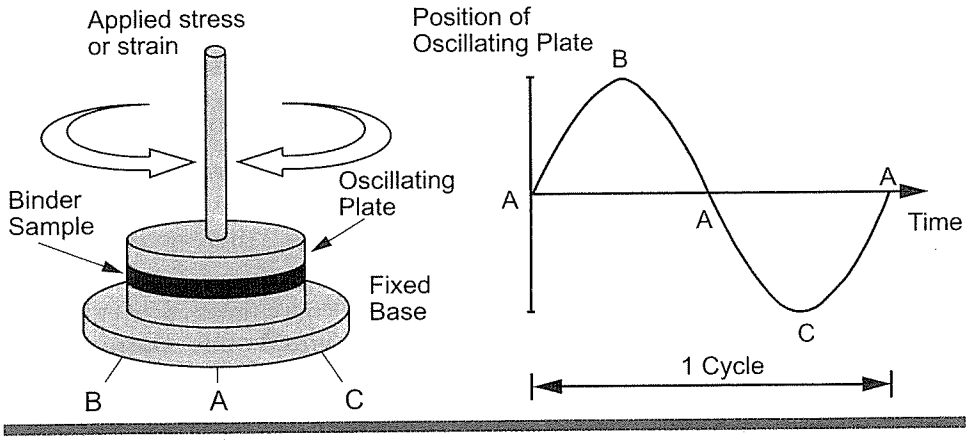


Figure 2.21 DSR Operation

point C. From point C the plate centerline moves back to point A. This oscillation is one cycle and is repeated for a specified number of cycles. The number of cycles completed in one second is the loading frequency. Typically, DSR tests that are run for specification purposes are performed at a frequency of approximately 1.59 hertz (1.59 cycles per second)—equivalent to 10 radians per second. Stress and strain are measured during each loading cycle and used to calculate complex shear modulus and phase angle.

To conduct testing, the DSR equipment (see Figure 2.22) is first initialized to operate with the proper parallel plate geometry and gap. The plate diameter and gap thickness used depend on the aged state of the asphalt being tested. Original (unaged) and short-term aged binders are tested with a 25-millimeter-diameter plate and a 1000-micron gap; long-term aged

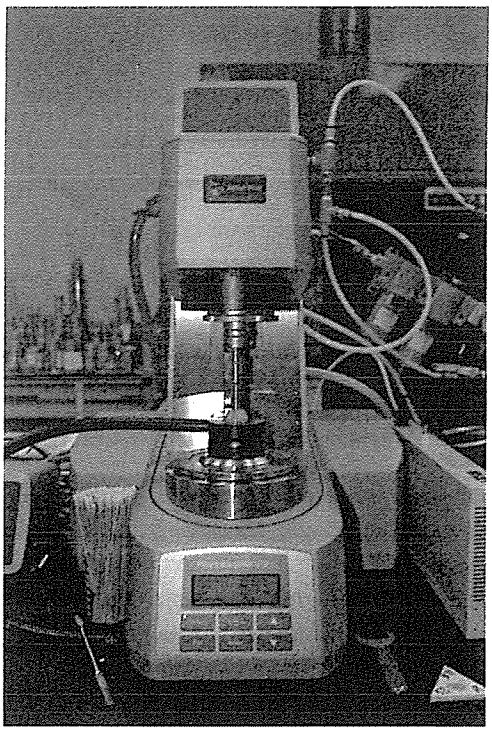


Figure 2.22 Dynamic Shear Rheometer

binder is tested with an 8-millimeter-diameter plate and a 2000-micron gap. Before mounting the specimen, the gap is set at the desired value (1000 or 2000 microns) plus an extra gap, usually 50 microns, to allow for the creation of a slight bulge in the test specimen.

To prepare the test specimen, the asphalt binder is heated until fluid, and is poured either directly onto one of the two parallel plates or into a silicone mold with the appropriate diameter and thickness for testing. After the asphalt binder is loaded, the parallel plates are brought together until achieving the appropriate gap setting (either 1000 or 2000) plus the extra gap amount (usually 50 microns). With the asphalt binder sample sandwiched between the plates, the sample is trimmed so that the asphalt binder is flush with the edges of the parallel plates. This trimming is needed to control the diameter of the test specimen and enable an accurate calculation of G^*. After trimming, the extra gap is eliminated so that the final gap, or specimen thickness, is exactly at the desired value. The specimen will bulge slightly as shown in Figure 2.23.

Because asphalt binder properties are temperature dependent, DSRs must have a precise means of controlling the sample temperature. This control is accomplished using either a circulating-fluid bath or a forced-air oven. In either case, the temperature must be controlled so that the sample temperature is uniform and varies by no more than 0.1°C across the gap.

Once the asphalt binder specimen is loaded and trimmed, the temperature of the test specimen needs time to equilibrate. While a minimum

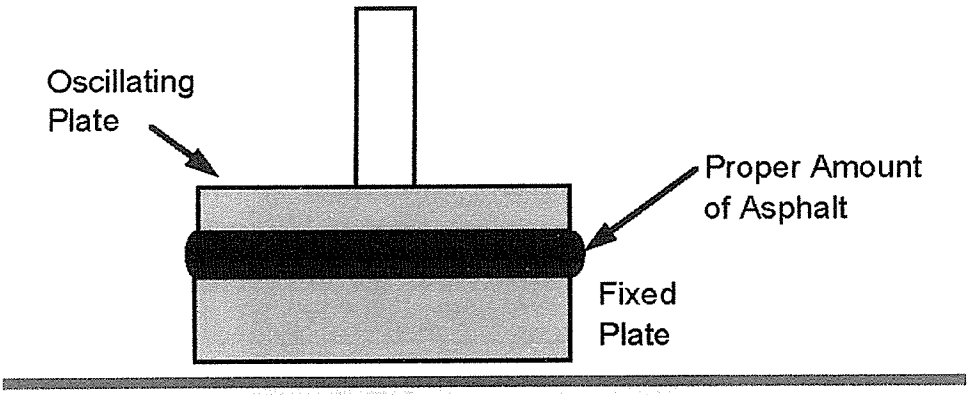


Figure 2.23 Asphalt Sample Configuration in DSR

equilibration time of 10 minutes is commonly used, the actual temperature equilibration time is equipment dependent and should be checked using a calibration specimen with very accurate temperature-sensing capabilities.

Before testing, the operator sets the approximate value of shear strain to be achieved during the test. Desired shear strain values vary from about 1 percent to 12 percent and depend on the aged state of the binder being tested. Original (unaged) binders and short-term aged binders are tested at strain values of approximately 10 to 12 percent. Long-term aged binders are tested at strain values of about 1 percent. In all cases, strain values must be small enough that the response of the binder (G^*) remains in the linear viscoelastic (LVE) range. As long as the test remains within the LVE range of the material, G^* will be virtually unaffected by changes in strain level.

Once the test begins, everything is computer controlled to provide the proper sinusoidal loading while measuring stress and strain responses. The sample is first conditioned by loading for ten cycles. During this conditioning period, the DSR measures the stress required to achieve the set shear strain and then precisely maintains this stress during the test. After the ten conditioning cycles, ten additional cycles are applied to obtain test data. The software automatically computes and reports G^* and δ, using the relationship between the applied stress and the resulting strain.

The complex shear modulus, G^*, is the ratio of total shear stress ($\tau_{max}-\tau_{min}$) to total shear strain ($\gamma_{max}-\gamma_{min}$). The time lag between the applied stress and the resulting strain (or applied strain and resulting stress) is related to the phase angle, δ. For a perfectly elastic material, an applied load causes an immediate response; thus, the time lag or phase angle is zero. A viscous material (such as hot asphalt at mixing temperatures) has a relatively large time lag between load and response; in this case, the angle approaches 90 degrees. Because asphalt binders are viscoelastic at normal pavement temperatures, they behave somewhere between the two extremes. These responses are illustrated in Figures 2.24 and 2.25.

The formulas used by the rheometer software to calculate τ and γ are shown in Figure 2.26. The formulas are not shown so that the user can actually calculate G^*—this is accomplished by the DSR operating software—but

to illustrate the importance of the specimen geometry. In the determination of G^*, the radius of the specimen is raised to the fourth power. Thus, errors in trimming the test specimen can rapidly impact the calculated G^* value.

Bending Beam Rheometer

Asphalt binders at low temperatures are too stiff to reliably measure rheological properties using the parallel-plate geometry of a conventional

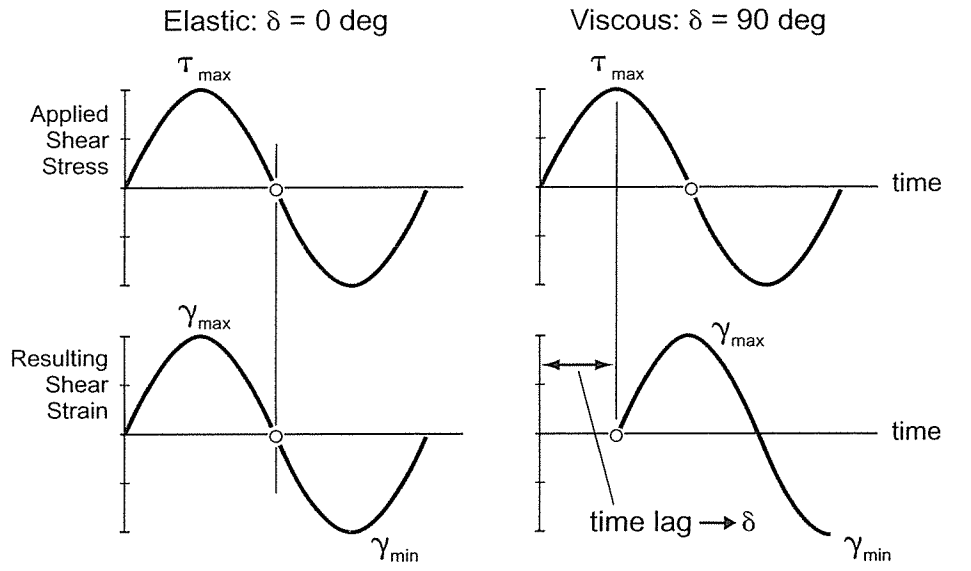


Figure 2.24 Stress-Strain Output

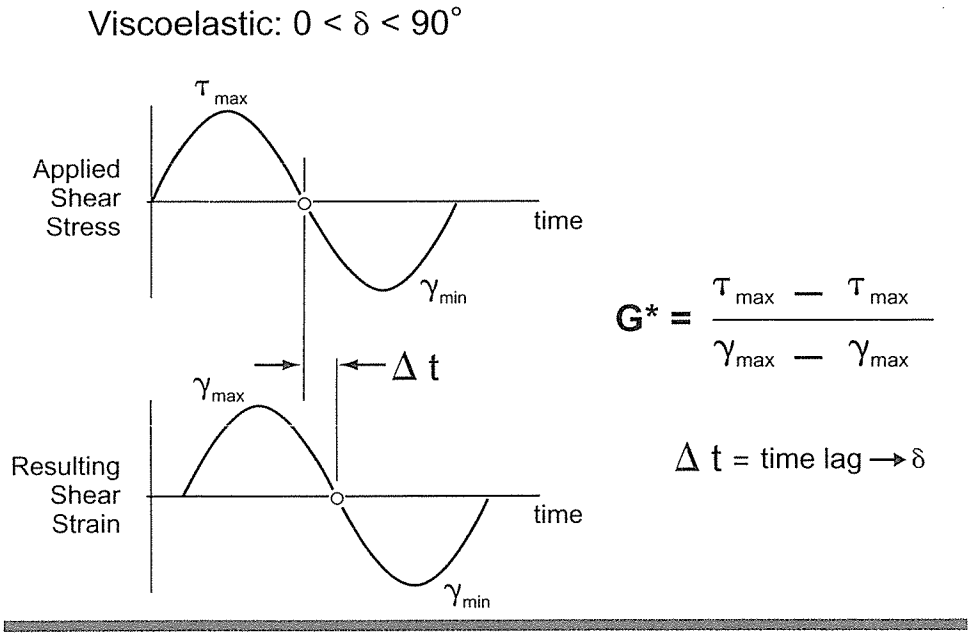
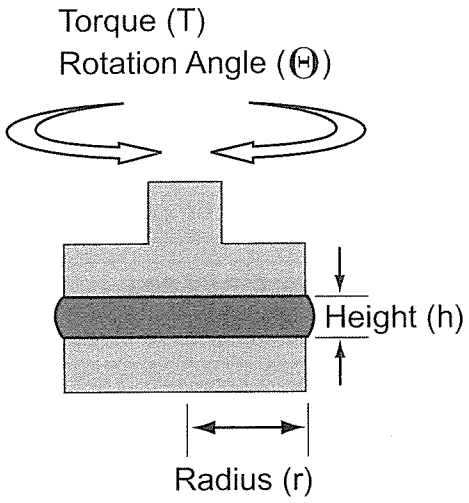


Figure 2.25 Stress-Strain Output of Viscoelastic Material



$$\tau = \frac{2T}{\pi r^3}$$

$$\gamma = \frac{\Theta r}{h}$$

where T = maximum applied torque,
r = radius of specimen (either 12.5 or 4 mm),
Θ = rotation angle, and
h = speciment height (either 1 or 2 mm)

Figure 2.26 Dynamic Shear Rheometer Calculations

DSR. Therefore, SHRP researchers developed the bending beam rheometer (BBR) to accurately evaluate binder properties at low pavement temperatures. Used together, the DSR and BBR tests provide stiffness behavior of asphalt binders over a wide range of temperatures.

The BBR is used to measure how much a binder deflects or creeps under a constant load at a constant temperature. BBR test temperatures are related to a pavement's lowest service temperature, when the asphalt binder acts more like an elastic solid than a viscous fluid. Furthermore, the test is performed on binders that have been aged to simulate hot mixing in a mixing facility and some in-service aging.

The BBR gets its name from the test specimen geometry and loading method used during testing. The key elements of the BBR (see Figure 2.27) are a loading frame, temperature-controlled fluid bath, and the computer-control, data-acquisition system. A blunt-nosed shaft applies a load to the midpoint of a simply supported asphalt binder beam. A load cell is mounted on the loading shaft, which is enclosed in an air bearing to eliminate friction during loading. Loads are applied by pneumatic pressure. A transducer on the loading shaft monitors deflection.

The BBR procedure uses beam theory to calculate the stiffness of an asphalt beam sample under a creep load. By applying a constant load to the asphalt beam and measuring the center deflection of the beam throughout the four-minute test procedure, the creep stiffness *(S)* and creep rate *(m-value)* can be calculated. *Creep stiffness* is the resistance of the asphalt binder to creep loading, and the *m-value* is the change in asphalt stiffness with time during loading.

The controlled temperature bath contains a fluid that will not freeze at the test temperatures used (potentially as low as –36°C). Fluids such as methanol, ethanol, ethylene glycol, water, and combinations of these are commonly used. As with the DSR, tight temperature control is needed (± 0.1°C).



Figure 2.27 Bending Beam Rheometer

Asphalt binder test specimens are prepared using a rectangular aluminum mold (see Figure 2.28). The mold assembly consists of an aluminum base, side and end plates, acetate base and side strips, and rubber O-rings. Before the mold is assembled, the inside surfaces of the two side plates and the base plate are lightly coated with a petroleum-based jelly. The three acetate strips are then placed against the greased faces. The end pieces are treated with a release agent composed of glycerin and talc. The mold is then assembled and held together with two rubber O-rings. The asphalt beam specimens are formed by heating the binder until fluid and filling the mold in a continuous motion from one end to the other.

After a cooling period of 45 to 60 minutes, excess asphalt is trimmed from the upper surface using a hot putty knife. The asphalt specimen remains in the mold at room temperature for a short time and is removed from the mold only when the conditioning procedure is ready to begin.

So that the asphalt binder test specimen can be removed from the mold, the assembly is cooled for 5 to 10 minutes in an ice bath or freezer. The mold is then rapidly disassembled and the asphalt binder beam is transferred immediately to the BBR bath for conditioning at the test temperature for 60 minutes.

After the 60-minute thermal conditioning period in the bath, the asphalt beam is loaded in the test frame and subjected to a constant creep load of 980 milliNewtons, for 240 seconds. Load and beam deflection are continuously measured during the test and are used to calculate creep stiffness *(S)* and creep rate (m-value) at specified load times—8, 15, 30, 60, 120, and 240 seconds—using beam theory. Figure 2.29 shows a sample graph and the procedure used to obtain the creep rate at each load time.

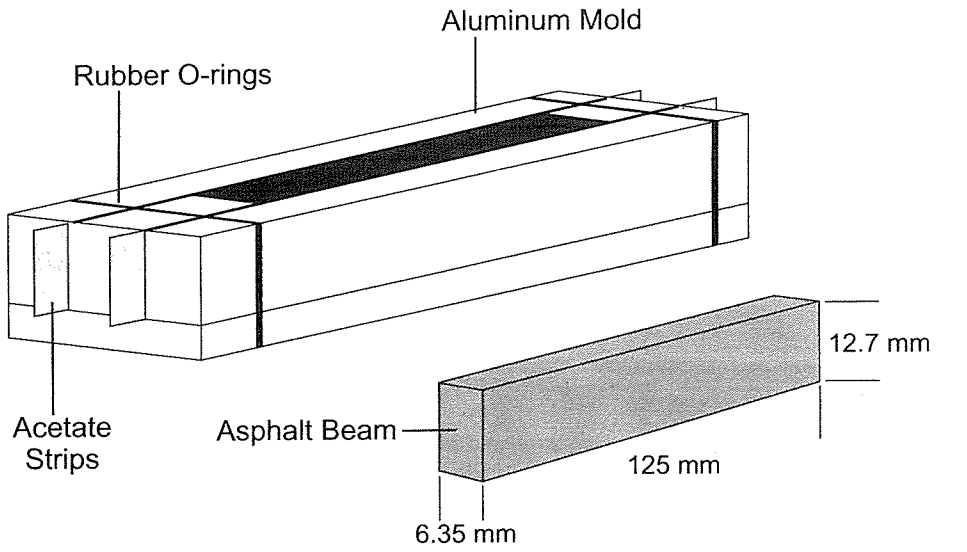


Figure 2.28 Bending Beam Rheometer Deflection (Mold Assembly)

To relate to field performance, the value of creep stiffness when the asphalt binder has been loaded for two hours at the minimum pavement design temperature should be determined. However, when using the concept of time-temperature superposition, SHRP researchers confirmed that by raising the test temperature 10°C, an equivalent creep stiffness can be obtained after only 60 seconds of loading. The obvious benefit is that a test result can be obtained in a much shorter testing time.

The second parameter determined from the results of the bending beam test is the m-value. The m-value represents the rate of change in the creep stiffness, *S(t)*, versus time. This value is also calculated automatically by the bending beam software. To obtain m-value, the stiffness is calculated at several loading times (see Figure 2.29). The m-value is the slope of the log stiffness versus log time curve at any time, *t*.

Computer-generated output for the bending beam test includes all the required reporting items. The output format generally includes plots of deflection and load versus time, actual load and deflection values at various times, test parameters, and operator information.

Direct Tension Tester

Numerous past studies have shown a strong relationship between the stiffness of conventional (unmodified) asphalt binders and the amount of stretching they undergo before breaking. Asphalts that undergo considerable stretching before failure are called ductile; those that break without much stretching are called brittle. It is important that an asphalt binder be capable of a minimal elongation. Typically, stiffer asphalts are more brittle and softer asphalts are more ductile.

Creep stiffness as measured by the BBR is not adequate to completely characterize the capacity of asphalts to stretch before breaking. For example,

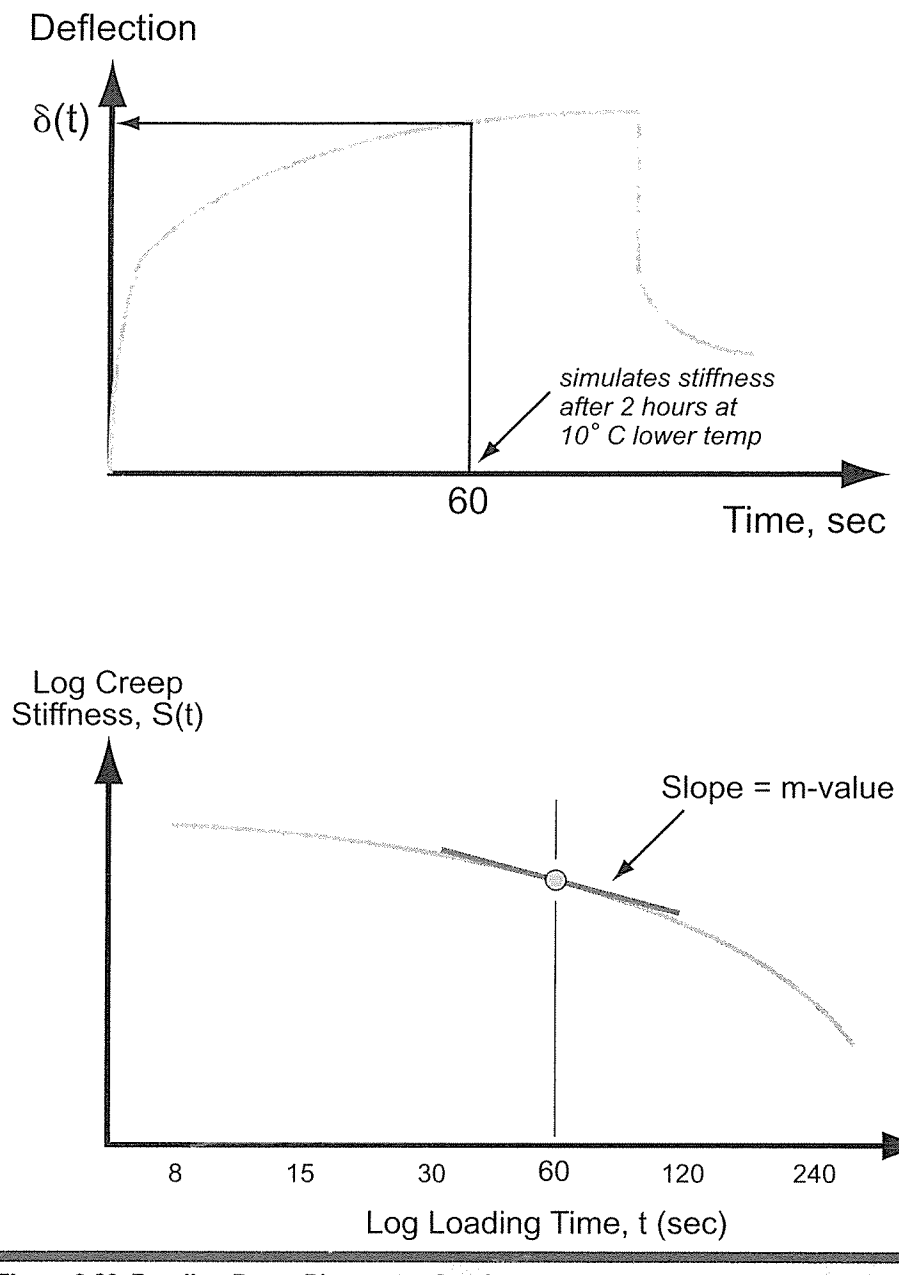


Figure 2.29 Bending Beam Rheometer Graph

some binders exhibit high creep stiffness but can also stretch farther before breaking. Consequently, SHRP researchers developed a test to evaluate the fracture properties of asphalt binders at low temperatures.

The direct tension test measures the amount of binder stress and strain at failure at very low temperatures. It is performed at temperatures where asphalt binders generally exhibit brittle behavior 0°C to –36°C. Furthermore, the test is performed on binders that have been subjected to long-term aging to simulate several years of in-service aging.

In the direct tension test, a small asphalt specimen shaped like a dog bone is pulled at a slow, constant rate until it fails. The elongation at failure is used to calculate the failure strain, which is an indication of whether the binder will behave in a brittle or ductile manner at the low test temperature.

The failure strain (ε_f) is calculated as the change in length divided by the original length, expressed as a percentage. In the direct tension test, failure is defined as the load when the stress reaches its maximum value, not necessarily the load when the specimen breaks. Failure stress (σ_f) is the failure load divided by the original cross section of the specimen.

In preparation for running the direct tension test, asphalt binder test specimens are formed in aluminum molds with end inserts, made from a phenolic resin and containing a hole inset with a stainless steel ring. As in the BBR test, heated asphalt binder is poured into the mold, filling it in a continuous motion from one end to the other. Six individual specimens are formed to produce an average test result.

After the specimens are poured, they are allowed to cool at ambient temperature for 30–60 minutes before trimming (see Figure 2.30). After trimming, the specimen is again allowed to cool for 10–15 minutes before it is carefully removed from the mold assembly. As in the BBR test, the test specimens are then subjected to temperature conditioning in the fluid bath at the specified test temperature for 60 minutes before testing.

Although the direct tension concept is simple, the equipment (see Figure 2.31) used requires complex features to accurately measure the very small

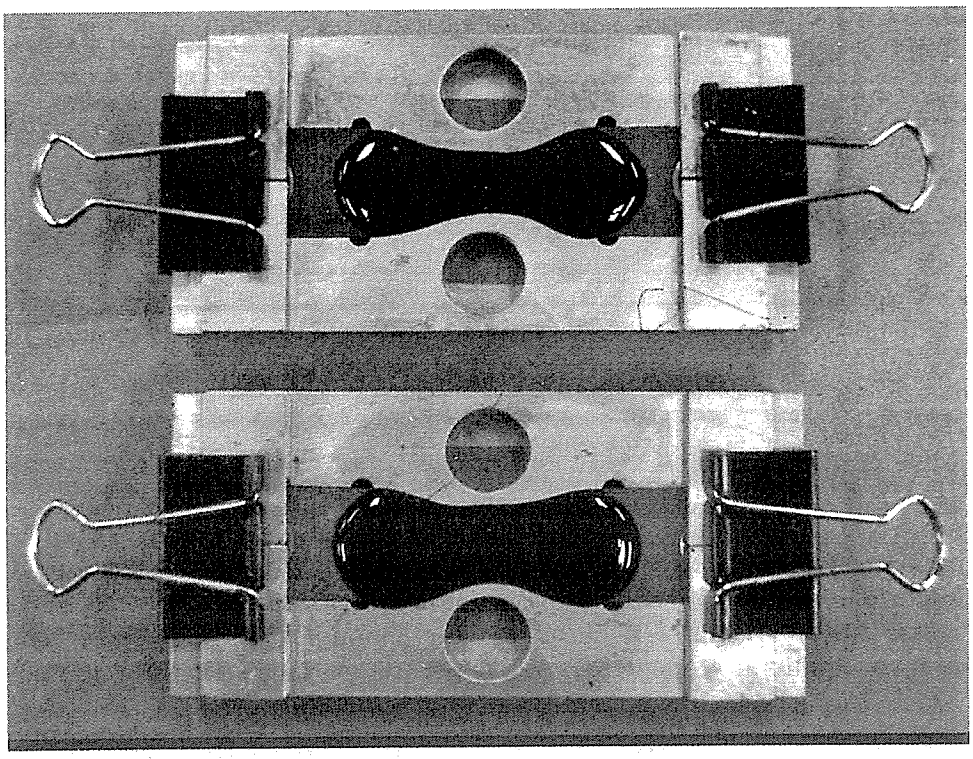


Figure 2.30 Direct Tension Specimens

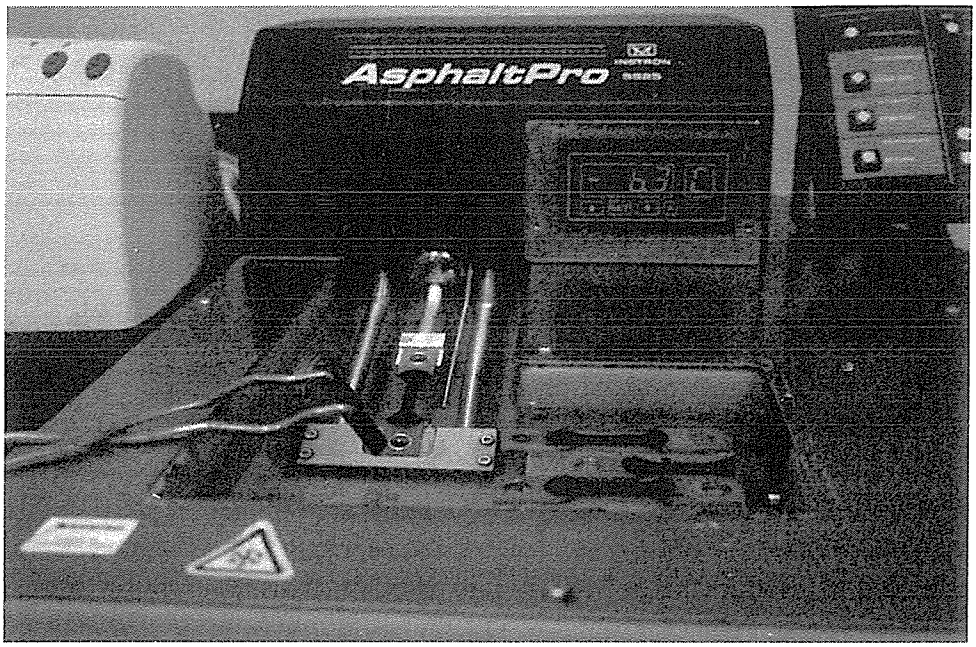


Figure 2.31 Direct Tension Tester

strains involved in the test. The direct tension tester (DTT) consists of three components: an electromechanical test device to apply load, a transducer system to measure and control grip separation, and an environmental control system (usually in the form of a controlled temperature liquid bath).

A key feature of the testing machine is the gripping system used to attach specimens to the alignment rods that apply the tensile load (see Figure 2.32). The grips have a ball-joint connection that ensures no bending within the binder test specimen.

The environmental control system includes an environmental chamber (or liquid bath) and a mechanical refrigeration unit capable of producing and precisely maintaining temperatures as low as –36°C. In the case of an environmental chamber, air is used as the cooling medium. If a liquid bath is used, a solution of potassium acetate and water is used as the cooling medium.

At the end of the conditioning period (60±5 minutes), each of the six specimens is tested individually. After mounting the specimen on the end grips, a tensile load is applied by pulling one end at a strain rate of 3 percent per minute (1.0 millimeter per minute), while the other end remains fixed, until the specimen fails. A test typically requires less than a minute from load application until failure. A test result is considered legitimate when fracture occurs within the midsection of the specimen with the constant cross section. A specimen that breaks near the end insert should be reported as an "end break."

After all six specimens have been tested (see Figure 2.33), the two lowest stress-at-failure values are eliminated and the remaining four values are

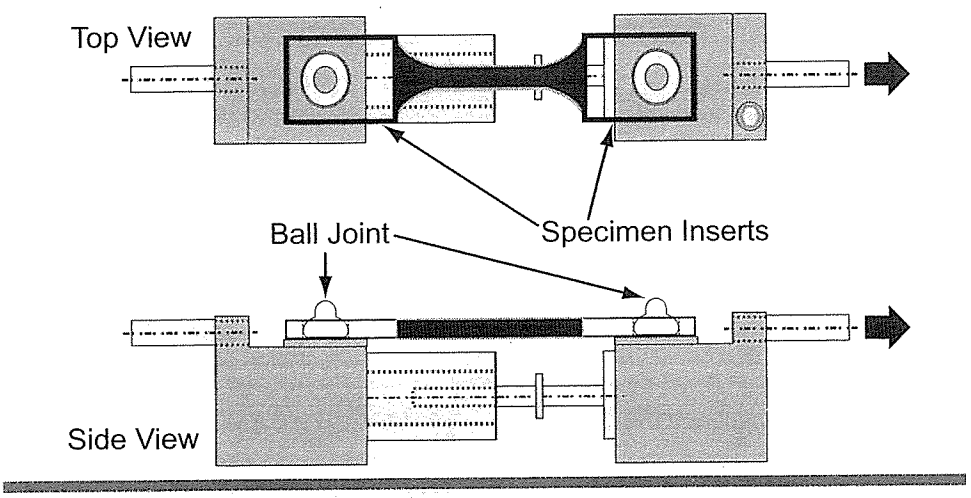


Figure 2.32 Direct Tension Loading

averaged to produce an average failure stress and average failure strain. This practice is based on the concept that it is much more common in direct tension tests to have an early fracture because of specimen or procedural imperfections than an exceptionally "strong" test specimen that does not fracture early. Other analysis procedures may dictate different methods for evaluating the quality of the test data and its suitability for inclusion in the general average.

A single test result consists of the average strain to failure of multiple DTT specimens, reported to the nearest 0.01 percent and the average stress at failure reported to the nearest 0.01 megaPascals.

Aging Procedures

Two aging procedures were adopted as part of the PG Asphalt Binder Specification system to address short-term and long-term aging. Rather than develop a new procedure for short-term aging, the SHRP research team elected to use the existing Rolling Thin Film Oven Test procedure. The details of the RTFOT are discussed in a previous section.

The RTFO-aged binder is used for DSR testing and is transferred into small containers for additional long-term aging and future use.

As described previously, the effects of long-term, in-service aging of asphalt were not incorporated into specifications for asphalt binders before the SHRP research and the PG Asphalt Binder Specification. To address this shortcoming, the SHRP research team used an existing research procedure and equipment, the Pressure Aging Vessel (PAV), and modified it to simulate long-term, in-service aging. The PAV exposes the asphalt binder to high pressure and temperature for 20 hours to simulate the effects of long-term aging. Because asphalt that has been exposed to long-term aging has also been through the mixing and construction process, the PAV procedure uses asphalt binder that has been previously aged in the RTFO.

The pressure aging apparatus consists of the pressure aging vessel within an oven to maintain temperature (see Figure 2.34). Air pressure is

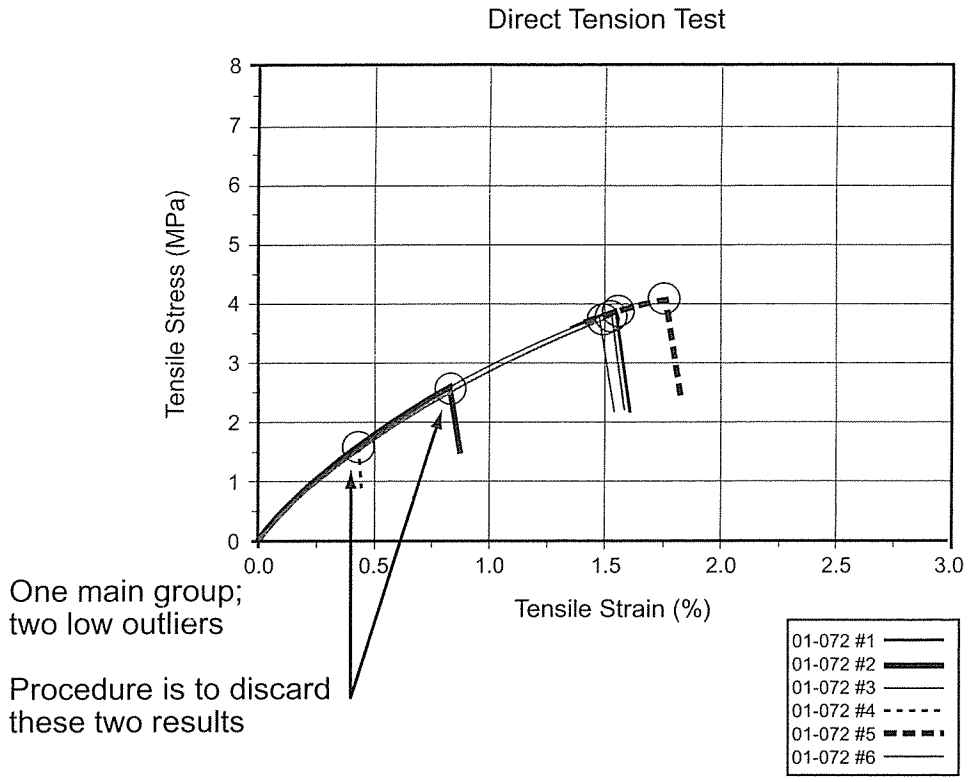


Figure 2.33 Stress-Strain Curves for Direct Tension

provided by a cylinder of clean, dry compressed air with a pressure regulator, release valve, and a slow-release bleed valve. The pressure vessel is designed to operate under the pressure (2.1 megaPascals) and temperature conditions 90°C, 100°C, or 110°C of the test. A sample rack, capable of holding ten sample pans, fits conveniently into the vessel.

To prepare for the PAV, RTFO-aged binder is heated until fluid and stirred to ensure homogeneity. Fifty grams of the heated asphalt binder are then poured into the PAV sample pan (see Figure 2.35), which is then loaded in the sample rack.

The unpressurized PAV is preheated to the desired test temperature. After preheating, the sample rack with the samples is placed in the hot vessel and the lid is quickly secured to avoid excessive heat loss. When the vessel temperature is within 2–20°C of the required temperature, the pressure is applied and the timing for the aging period begins.

After 20 hours, the pressure is gradually released, requiring 8 to 10 minutes to minimize the potential for sudden depressurization and foaming. The sample rack is then removed from the PAV. Sample pans are removed from the rack and placed in an oven at 163°C for 15 minutes before being scraped into a separate container for each asphalt binder (more than one pan may be aged for each asphalt binder) and being subjected to a vacuum

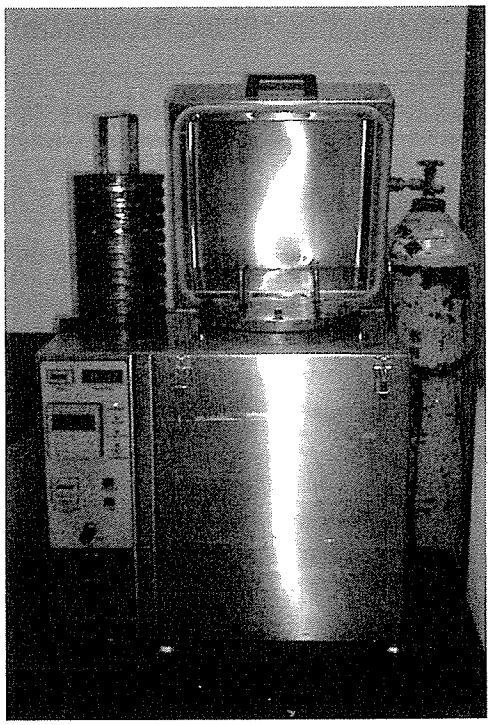


Figure 2.34 Pressure Aging Vessel (PAV)

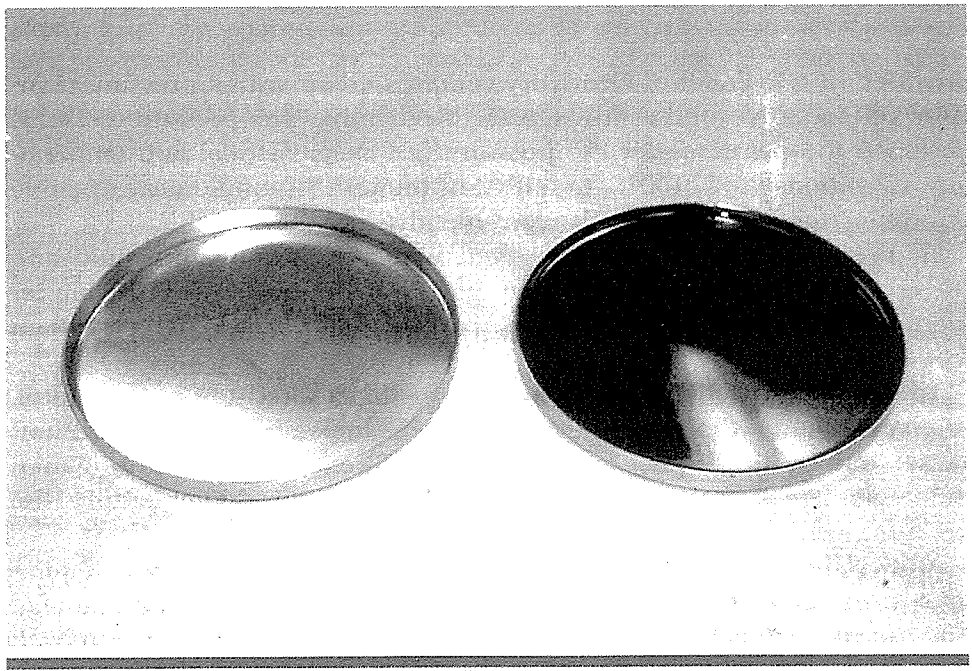


Figure 2.35 Pressure Aging Vessel Pans

degassing procedure (conducted in a 170°C vacuum oven for 30 minutes at a pressure of 15 kiloPascals absolute) to remove air bubbles from the aged asphalt binder sample.

The PAV procedure does not produce a test result; it is merely a long-term aging procedure. Asphalt binder that has been PAV-aged is used in dynamic shear rheometer, bending beam rheometer, and direct tension tests.

2.3.4 PG Asphalt Binder Specification

Introduction

The central theme of the PG Asphalt Binder Specification is its reliance on testing asphalt binders in conditions that simulate critical stages during the life of the asphalt binder. Tests performed on the original asphalt binder represent the first stage of transport, storage, and handling. The second stage represents the asphalt binder during mix production and construction and is simulated for the specification by aging the asphalt binder using the rolling thin film oven. This procedure exposes thin binder films to heat and air and approximates the aging of the asphalt during mixing and construction. The third stage occurs as the asphalt binder ages over a long period as part of the hot mix asphalt pavement layer. This stage is simulated for the specification by aging using the pressure-aging vessel. This procedure exposes asphalt binder samples to heat and pressure in order to simulate years of in-service aging in a pavement.

The PG Asphalt Binder Specification and the test methods used to characterize asphalt binders are described in AASHTO M 320 (see Figure 2.36). Table 1 of the specification provides requirements for PG asphalt binders using the previously described set of tests and aging procedures. Original, non-aged asphalt binder tests include the flash point, rotational viscosity, and DSR tests. RTFO aging is conducted to determine mass loss and provide short-term aged asphalt binder for another high temperature DSR test and additional aging. PAV aging is conducted to provide long-term aged asphalt binder for intermediate temperature DSR and low temperature BBR tests.

While the direct tension test is rarely used in Table 1, it is an integral part of Table 2 requirements. In AASHTO M 320, Table 2 requirements are identical to Table 1 requirements except for low temperature testing. In Table 2, an alternate procedure for determining the critical low cracking temperature of an asphalt binder is used that combines results from the bending beam rheometer and direct tension test procedures. Some researchers believe that the alternate approach in Table 2 is more rigorous and inclusive of most modified asphalts.

Advantages Compared with Older Specifications

A unique feature of the PG Asphalt Binder Specification is that, for the DSR and BBR tests, the specified criteria remain constant, but the temperature at which the criteria must be achieved changes for the various grades.

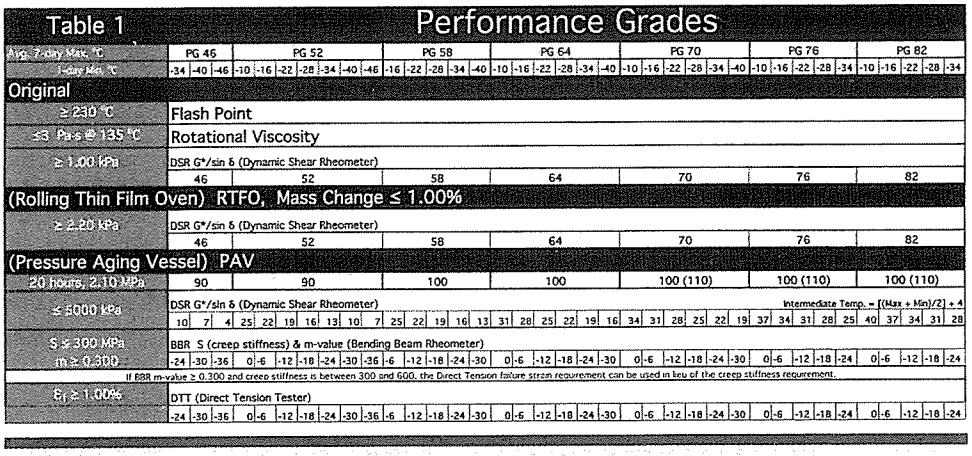


Table 1	Performance Grades						
Avg. 7-day Max. °C	PG 46	PG 52	PG 58	PG 64	PG 70	PG 76	PG 82
1-day Min. °C	-34 -40 -46	-10 -16 -22 -28 -34 -40 -46	-16 -22 -28 -34 -40	-10 -16 -22 -28 -34 -40	-10 -16 -22 -28 -34 -40	-10 -16 -22 -28 -34	-10 -16 -22 -28 -34
Original							
≥ 230 °C	Flash Point						
≤3 Pa·s @ 135 °C	Rotational Viscosity						
≥ 1.00 kPa	DSR G*/sin δ (Dynamic Shear Rheometer)						
	46	52	58	64	70	76	82
(Rolling Thin Film Oven) RTFO, Mass Change ≤ 1.00%							
≥ 2.20 kPa	DSR G*/sin δ (Dynamic Shear Rheometer)						
	46	52	58	64	70	76	82
(Pressure Aging Vessel) PAV							
20 hours, 2.10 MPa	90	90	100	100	100 (110)	100 (110)	100 (110)
≤ 5000 kPa	DSR G*/sin δ (Dynamic Shear Rheometer)						Intermediate Temp. = [(Max + Min)/2] + 4
	10 7 4	25 22 19 16 13 10 7	25 22 19 16 13	31 28 25 22 19 16	34 31 28 25 22 19	37 34 31 28 25	40 37 34 31 28
S ≤ 300 MPa m ≥ 0.300	BBR S (creep stiffness) & m-value (Bending Beam Rheometer)						
	-24 -30 -36	0 -6 -12 -18 -24 -30 -36	-6 -12 -18 -24 -30	0 -6 -12 -18 -24 -30	0 -6 -12 -18 -24 -30	0 -6 -12 -18 -24	0 -6 -12 -18 -24
If BBR m-value ≥ 0.300 and creep stiffness is between 300 and 600, the Direct Tension failure strain requirement can be used in lieu of the creep stiffness requirement.							
ε_f ≥ 1.00%	DTT (Direct Tension Tester)						
	-24 -30 -36	0 -6 -12 -18 -24 -30 -36	-6 -12 -18 -24 -30	0 -6 -12 -18 -24 -30	0 -6 -12 -18 -24 -30	0 -6 -12 -18 -24	0 -6 -12 -18 -24

Figure 2.36 Perfomance Graded Asphalt Binder Specification Table

As an example, consider two construction projects—one at the equator and one at the Arctic Circle. Good performance is expected in both locations, but the temperature conditions under which specified asphalt binder properties must be achieved are vastly different. This approach differs from previous specification systems, which relied on measuring a property at a fixed temperature but allowed the criteria to change based on the climate in which the asphalt binder would be used.

The PG asphalt binder tests measure physical properties that can be related directly to field performance by engineering principles. The use of complex shear modulus at high test temperatures allows for a better characterization of both the elastic and viscous components of the asphalt binder. This makes the test more appropriate for modified asphalt binders—many of which have a substantial elastic component at high temperature—than the absolute viscosity test.

Unlike previous specifications, the PG asphalt binder tests are conducted at the entire temperature range expected to be encountered by the asphalt binder during production and in-service in the asphalt pavement. Previous systems only routinely characterized physical properties as low as 77°F (25°C), with a few tests conducted at 41°F (4°C). Because asphalt pavements are frequently exposed to much lower temperatures, the ability to measure physical properties at a low temperature using the BBR and direct tension tests is a significant advancement.

Finally, the adoption of the pressure aging vessel procedure to simulate the long-term aging of asphalt binders is also an advancement in the characterization of asphalt binder properties. Fatigue and thermal cracking are most likely to occur in an asphalt pavement after the asphalt binder has been in service for several years. The pressure aging vessel procedure simulates this in-service aging, allowing for testing to be conducted at intermediate and low temperatures to determine the physical properties of the asphalt binder believed to be related to these distresses.

Assumptions in the Specification

The PG Asphalt Binder Specification incorporates several simplifying assumptions that permit the use of properties such as complex shear modulus (G^*). Asphalt binders that violate these assumptions cannot be effectively graded in the PG Asphalt Binder Specification. The asphalt binders must be rheologically simple systems.

First, the asphalt binders should be homogeneous. Particulate matter that has a particle size greater than 0.25 millimeter may interfere with the consistent response of tests, such as the DSR test at a high temperature. The size of such particles may cause the G^* measurement to be inaccurate and display behavior that is not linear viscoelastic. Asphalt binders with particulate additives may not meet this assumption.

Although not listed in Table 1 or Table 2, the asphalt binder must have a solubility of 99 percent or more according to AASHTO M 320. This requirement helps to ensure that excess inorganic material is not present in the asphalt binder.

Second, the asphalt binders should exhibit isotropic behavior. Isotropic behavior occurs when specimen loading or particle orientation has no effect on the response. Asphalt binders that incorporate fibers could exhibit anisotropic behavior—meaning the fiber orientation affects the test response.

Third, PG asphalt binders should exhibit linear viscoelastic behavior over a wide range of strain. Linear viscoelastic behavior can be determined by performing a strain sweep with the DSR. In this procedure, the asphalt binder sample is tested at several increasing strain levels with G^* determined at each strain level. To be a linear viscoelastic material, G^* should remain relatively constant over a wide range of shear strain.

How the Specification Addresses Pavement Performance

Safety

In the PG specification, safety is considered by using an older asphalt binder test—the Cleveland Open Cup Flash Point test. This test procedure (AASHTO T 48) indicates the temperature at which an asphalt binder will instantaneously flash in the presence of an open flame. The flash point is well below the fire point, the temperature at which the asphalt binder will actually burn.

The minimum temperature required for all PG asphalt binders is 230°C. This test is performed on the unaged binder.

Pumping and Handling

To ensure that asphalt binders, especially modified asphalts, can be pumped and handled at the hot mixing facility, the PG specification contains a maximum viscosity requirement on the non-aged binder as determined by AASHTO T 316. This value is 3 Pascal-seconds and it must be achieved at 135°C for all grades. The specifying agency may waive this requirement if the supplier warrants that the binder can be handled and pumped at the necessary temperatures.

While the limiting viscosity of 3 Pascal-seconds is not a problem for most PG asphalt binders, it may be exceeded for some highly modified asphalt binders. Users requesting an asphalt binder having a temperature spread (high-temperature grade to low-temperature grade) of greater than 100 degrees should expect the increased likelihood that the supplied asphalt binder will exceed the rotational viscosity limit of 3 Pascal-seconds (3000 centipoise) at 135°C. Exceeding the limit in these instances should not be considered a material property failure, but simply a result of the high modification level needed to achieve the required temperature spread.

The user agency, HMA contractor, and asphalt binder supplier should work together to ensure that the required grade is appropriate for the project and that the supplied asphalt binder can be properly handled by the HMA contractor to produce a mixture having the desired in-place properties.

Permanent Deformation

As discussed in the section describing the DSR, the total response of an asphalt binder to load consists of elastic (recoverable) and viscous (nonrecoverable) components. Pavement rutting, or permanent deformation, is the accumulation of nonrecoverable deformation in the asphalt mixture in response to repeated load applications at high temperatures. To address the response of the asphalt binder to these repeated loads, the PG specification defines and places requirements on a rutting factor, $G^*/\sin\delta$ (read as "G star over sine delta"), which represents a measure of the high temperature stiffness of the asphalt binder. This factor is determined by dividing the complex shear modulus (G^*) by the sine of the phase angle (δ), as determined using the dynamic shear rheometer test procedure (AASHTO T 315). To minimize the contribution of the asphalt binder to rutting, the value of $G^*/\sin\delta$ must be a minimum of 1.00 kiloPascal for the original asphalt binder and 2.20 kiloPascals after aging the asphalt binder using the RTFO procedure at the appropriate grade temperature.

A minimum RTFO $G^*/\sin\delta$ value ensures that the asphalt binder will have sufficient stiffness after construction to minimize the contribution of the asphalt binder to permanent deformation in the asphalt mixture. High values of G^* and low values of δ are considered desirable attributes from the standpoint of rutting resistance. Thus, the PG specification promotes the use of stiff, elastic binders (non-aged and RTFO-aged) to address permanent deformation.

Excessive Aging

As with previous grading systems, a mass loss requirement is specified to help prevent the use of an asphalt binder that would age excessively from volatilization during hot mixing and construction. The mass loss requirement is calculated using the RTFO procedure (AASHTO T 240). The mass loss for any PG asphalt binder must not exceed 1.00 percent.

Fatigue Cracking

Like permanent deformation, G^* and δ are also used in the PG specification to address the response of the asphalt binder to the cumulative